

CellState Agent

Biological Report

GZMB CD8 T cells in normal bone marrow versus newly diagnosed multiple myeloma

Exploratory cross-dataset pseudobulk differential expression with dataset-level leave-one-out robustness and signed TF activity

Central biological conclusion

NDMM-associated GZMB CD8 T cells show a coherent interferon-stimulated and inflammatory state marked by IFI27, IFI44L, IFIT1/2/3, KLRK1, TNF, IFNG, PDCD1 and TIGIT. This is not a uniform increase in cytotoxicity: GZMB, CCL5 and FGFBP2 are not significantly higher in NDMM, while PRF1, GNLY, NKG7 and CTSW are modestly higher in NBM. The NBM-associated signature also contains differentiation-preserving T-cell genes together with strong myeloid, antigen-presentation and erythroid transcripts, indicating substantial substate-composition or technical-lineage sensitivity.

167,355	288	3,320	2,395
contributing cells	biological pseudobulk replicates	FDR-significant genes	LODO-conserved genes

Interpretation boundary

The contrast is NBM - NDMM: positive fold changes are higher in NBM and negative fold changes are higher in NDMM. No dataset contains both groups, so group and dataset are not independently identifiable. The report therefore describes robust atlas-associated state differences, not a dataset-adjusted causal disease effect.

Evidence bundle e70fcedaf8e1250049a18c0560d9572dd80f8e1c4abe9705b1c517b23e3d09a5
Unit of inference: independent biological pseudobulk replicate | DESeq2 design: ~ group | Minimum 100 cells per replicate
Scientific values preserved; sample identifiers and private paths sanitized in the public evidence export.

Executive interpretation

The dominant interpretable NDMM signal is an interferon-exposed, inflammatory and inhibitory-receptor-bearing state. IFI27, IFI44L, IFIT1, IFIT2, IFIT3, MX1, XAF1, OASL, GBP1 and ISG15 are higher in NDMM, together with KLRK1, TNF, IFNG, PDCD1 and TIGIT. Most of the strongest interferon and inflammatory genes retain the same direction in every dataset-level leave-one-out fold.

The contrast does not support the simpler claim that NDMM GZMB CD8 T cells are globally more cytotoxic. GZMB itself is not significant (log2FC 0.20, FDR 0.068), and CCL5 and FGFBP2 are also unchanged. PRF1, GNLY, NKG7 and CTSW are modestly higher in NBM. The data instead support qualitative remodeling of the GZMB CD8 compartment toward interferon signaling, inflammatory cytokine expression and checkpoint adaptation in the NDMM-associated atlas state.

The NBM-associated side is biologically mixed. CCR7, SELL, TCF7, LEF1, BACH2 and FOXO1 support a differentiation-preserving or memory-like component, while TOX, HAVCR2, ENTPD1, NR4A2 and CD69 indicate chronic stimulation or activation. In parallel, FCER1G, TYROBP, LYZ, S100A8/A9/A12, CST3, HLA-DRA, CD74, HBB, AHSP, ALAS2, GATA1 and SPI1 form a strong non-T-cell block. This pattern is more consistent with cell-state heterogeneity, ambient RNA, doublets or dataset-specific annotation/preprocessing than with a single coherent CD8 T-cell program.

Evidence hierarchy

Level	Defensible claim	Boundary
High within bundle	A coherent NDMM-associated interferon/inflammatory program is present and largely LODO-stable.	Atlas association; not independent of dataset.
Moderate	NDMM GZMB CD8 T cells show checkpoint adaptation, while NBM contains differentiation-preserving and chronic-stimulation substates.	May reflect shifts in substate composition within the broad label.
Low / QC-sensitive	SPI1 and other NBM-positive TF scores reflect a lineage-sensitive regulatory block.	TF activity is model-based and strongly affected by mixed-lineage targets.
Not established	Disease causality, antigen specificity, exhaustion as a formal state, or therapeutic tractability.	Requires balanced cohorts and orthogonal validation.

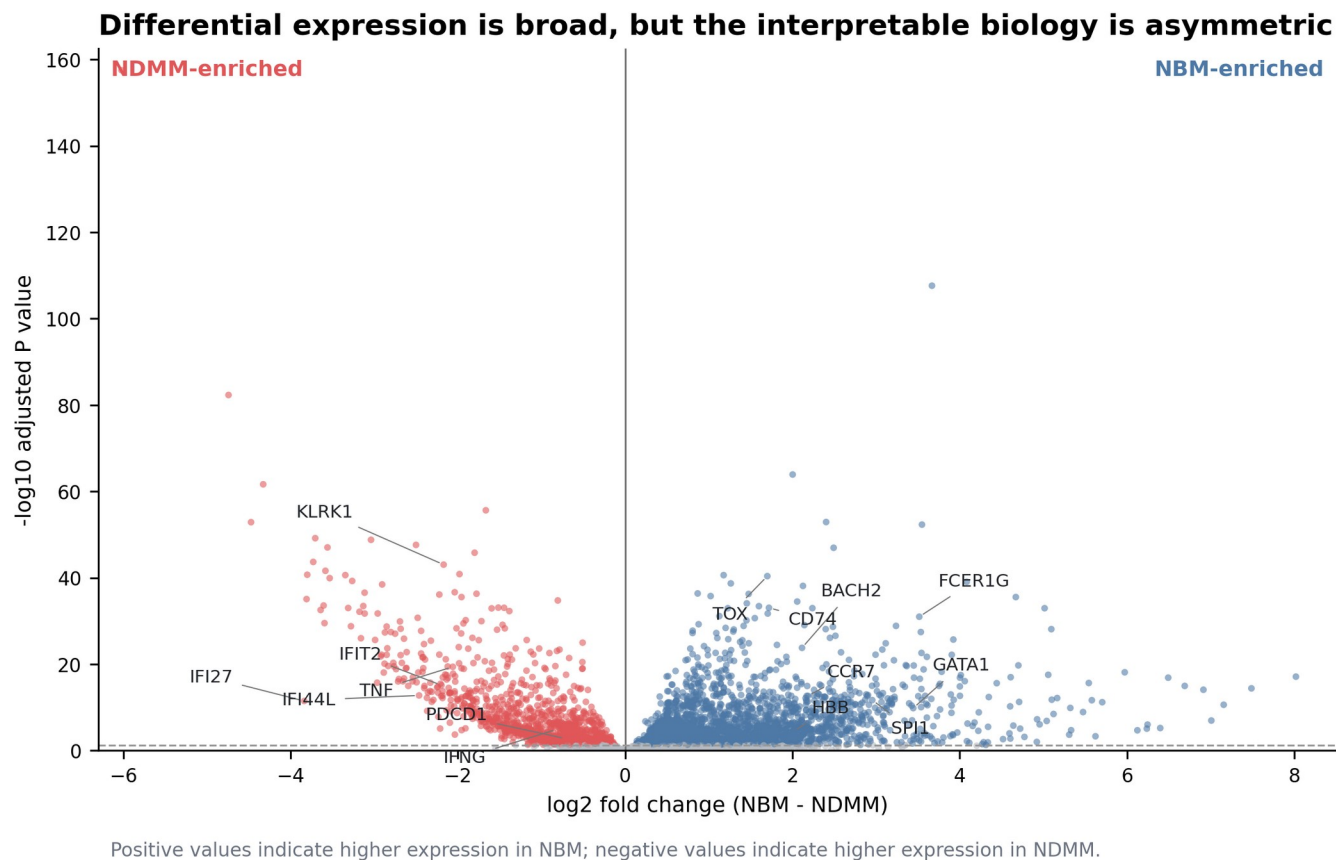


Figure 1. The broad differential-expression landscape contains a coherent NDMM interferon/inflammatory block and a biologically mixed NBM block. Positive effects are higher in NBM; negative effects are higher in NDMM. Evidence: deseq2_results.tsv

1. NDMM-associated GZMB CD8 T cells are interferon-exposed and inflammatory

The most internally coherent disease-associated program is the interferon-stimulated block. IFI27 (log2FC -3.84), IFI44L (-2.46), IFIT2 (-2.20), IFIT1 (-2.08), IFIT3 (-1.82), XAF1 (-1.31), MX1 (-1.10), OASL (-0.78), GBP1 (-0.71) and ISG15 (-0.73) are all higher in NDMM. IFI27, IFI44L, IFIT1/2/3, XAF1, MX1, OASL and GBP1 satisfy the predefined LODO conservation rule; ISG15 remains NDMM-directed in nine of ten folds but weakens when MMRF_ImmuneAtlas is omitted.

This interferon program is accompanied by inflammatory effector and receptor remodeling. KLRK1/NKG2D (-2.17) and TNF (-2.06) are among the strongest coding-gene effects and are conserved across all folds. IFNG (-0.81), PDCD1 (-0.70), TIGIT (-0.42) and CX3CR1 (-1.00) are also higher in NDMM. PDCD1 is conserved across all folds; IFNG and TIGIT reverse direction only in the fold omitting MMRF_ImmuneAtlas, indicating that the largest NDMM cohort materially contributes to those two effects.

Biological interpretation

The joint presence of interferon-stimulated genes, TNF/IFNG, KLRK1 and inhibitory receptors is consistent with an activated, cytokine-exposed and adaptively restrained GZMB CD8 state in the NDMM marrow environment. The data do not establish antigen specificity or a formal exhaustion phenotype, but they do support chronic immune stimulation and checkpoint engagement as testable hypotheses.

Marker-defined programs resolve a coherent NDMM inflammatory state

Negative values: higher in NDMM Positive values: higher in NBM

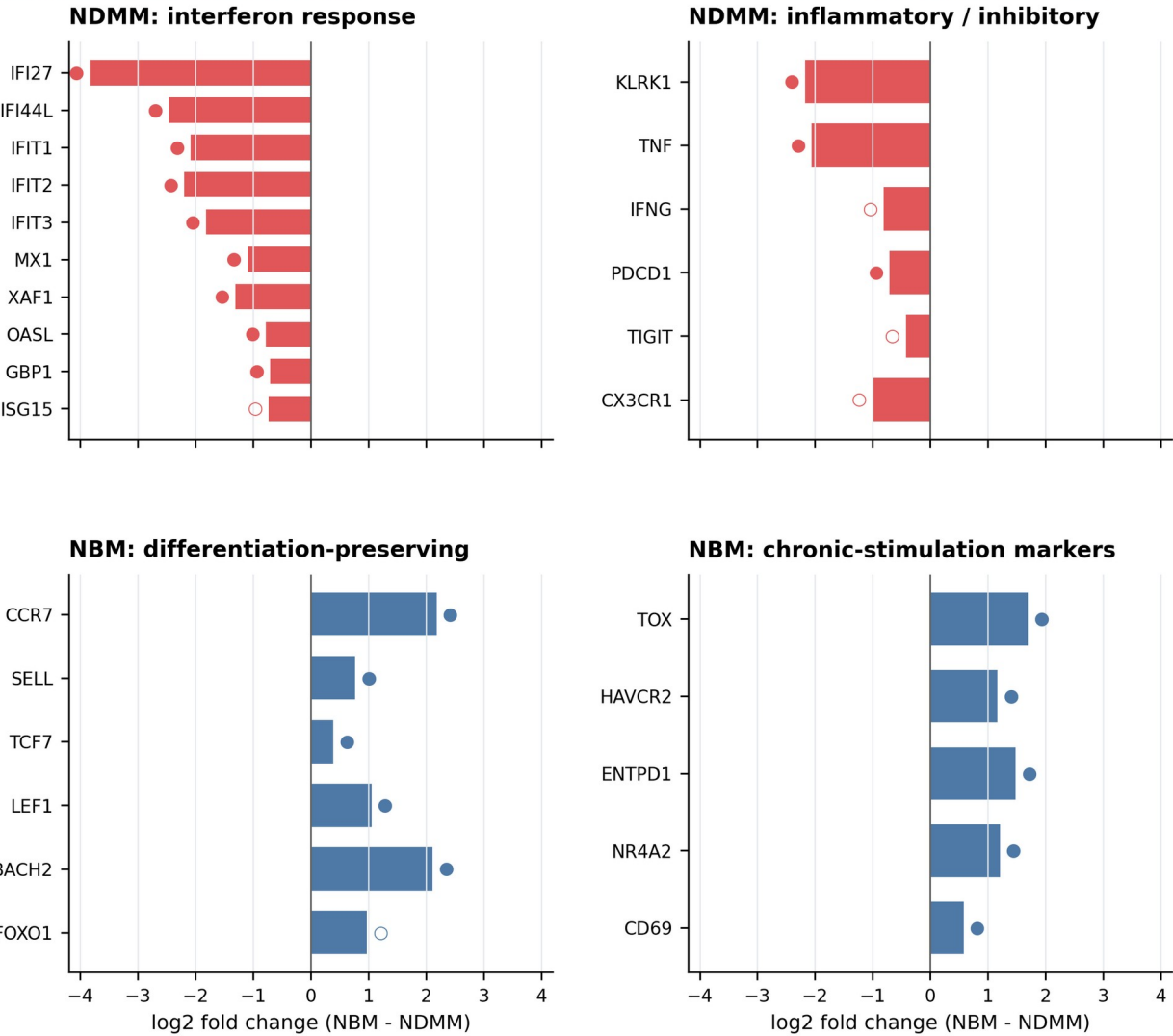


Figure 2. Curated marker panels expose the biological structure hidden by the large number of significant genes. Filled circles indicate genes meeting the predefined LODO conservation rule; open circles remain directionally stable but fail at least one conservation threshold. Marker panels are interpretive summaries, not formal pathway-enrichment tests. Evidence: deseq2_results.tsv; lodo_feature_summary.csv

Representative NDMM-associated genes

Gene	log2FC	FDR	LODO direction	Interpretation
IFI27	-3.84	3.3e-12	10/10	interferon response
IFI44L	-2.46	1.7e-13	10/10	interferon response
IFIT2	-2.20	7.2e-16	10/10	interferon response
KLRK1	-2.17	8.9e-44	10/10	NKG2D receptor
TNF	-2.06	3.9e-20	10/10	inflammatory cytokine

IFNG	-0.81	1.1e-05	9/10	type II cytokine
PDCD1	-0.70	0.002	10/10	inhibitory receptor
TIGIT	-0.42	0.011	9/10	inhibitory receptor

2. The disease contrast is not a simple cytotoxicity axis

The broad cell-state label is GZMB CD8 T cell, but canonical cytotoxic genes do not move as a single module. GZMB, CCL5 and FGFBP2 are not differentially expressed at FDR < 0.05. PRF1 (+0.34), GNLY (+0.36), NKG7 (+0.22) and CTSW (+0.92) are modestly higher in NBM. This argues against describing the NDMM state as simply more cytotoxic or more terminally differentiated.

The NBM-associated T-cell component instead combines differentiation-preserving genes - CCR7 (+2.19), SELL (+0.77), TCF7 (+0.39), LEF1 (+1.05), BACH2 (+2.11) and FOXO1 (+0.98) - with chronic-stimulation markers TOX (+1.70), HAVCR2 (+1.17), ENTPD1 (+1.48), NR4A2 (+1.21) and CD69 (+0.58). These apparently opposing programs imply heterogeneity within the GZMB CD8 label and likely differences in the proportions of constituent substates between the assembled NBM and NDMM cohorts.

The strongest concern is the NBM-associated mixed-lineage block. FCER1G (+3.52), TYROBP (+1.73), LYZ (+1.35), S100A8 (+2.17), S100A9 (+2.05), S100A12 (+2.41), CST3 (+2.11), CD74 (+1.72), HBB (+2.00), AHSP (+2.40), ALAS2 (+2.46), GATA1 (+3.44) and SPI1 (+2.95) are difficult to reconcile as a cell-intrinsic CD8 T-cell program. Their persistence across LODO folds means they are systematic within the assembled bundle, not necessarily biological.

The contrast is not a simple gain of cytotoxicity in NDMM

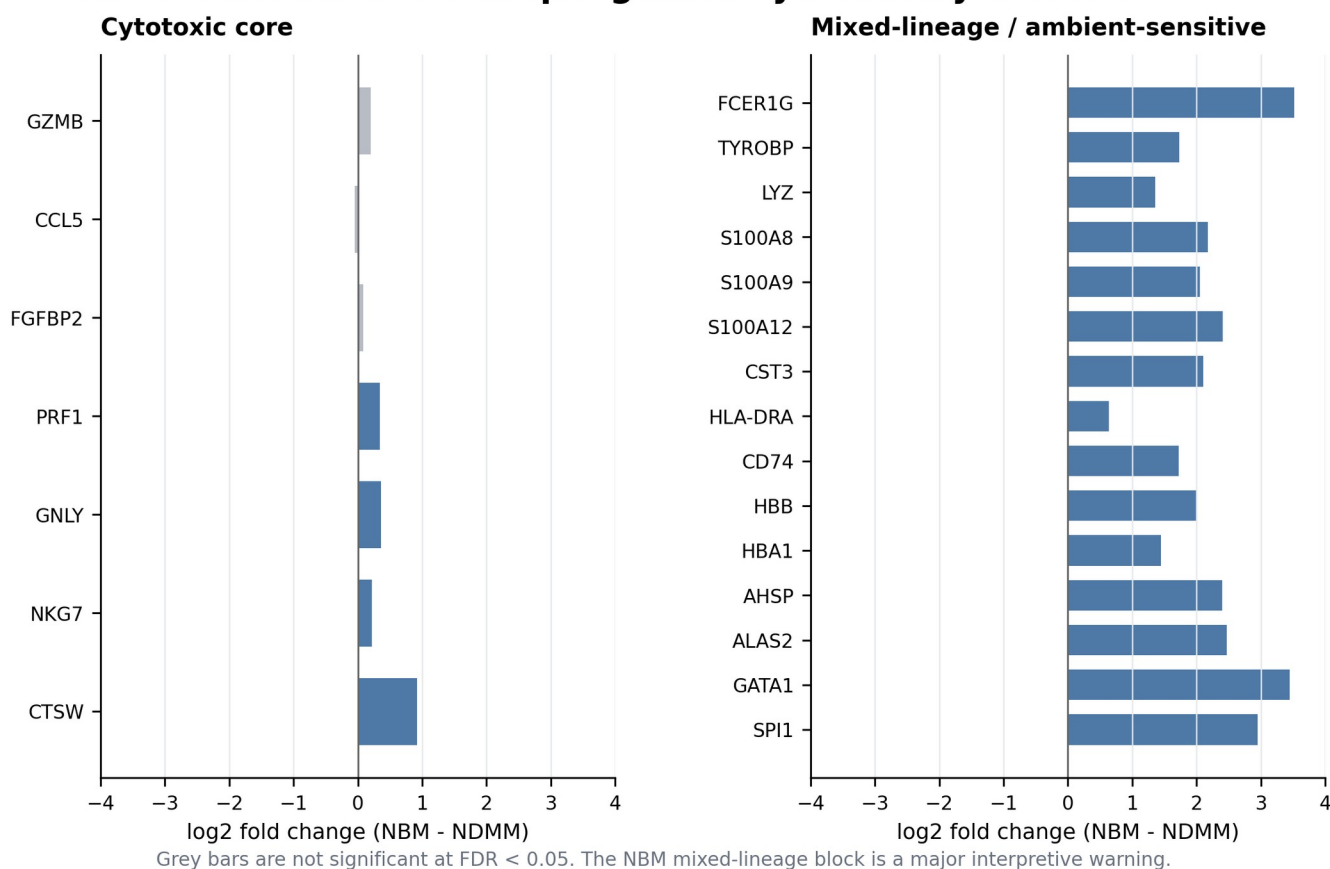


Figure 3. Cytotoxic-core genes do not support a global NDMM gain, whereas myeloid, antigen-presentation and erythroid transcripts form a strong NBM-associated block. Grey bars are not significant at FDR < 0.05. Evidence: deseq2_results.tsv

Most likely explanations for the NBM mixed-lineage block

Residual ambient RNA, doublets, differences in cell-state annotation granularity, marrow processing, chemistry or quality-control thresholds, and shifts in the composition of cytotoxic CD8 substates are all plausible. The current bundle cannot distinguish these mechanisms because the groups are segregated by dataset.

3. Leave-one-dataset-out robustness preserves the core signal

All 10 of 10 dataset-level LODO folds were estimable, and 2,395 genes met the predefined conservation rule. The rule required a significant full-data effect, at least 80% directional consistency, a median absolute LODO effect of at least 0.25, and no opposite-direction fold under the configured threshold.

The central NDMM interferon program is highly stable: IFI27, IFI44L, IFIT2, KLRK1, TNF and PDCD1 remain NDMM-associated in every omitted-dataset analysis. The NBM differentiation block - CCR7, SELL, TCF7, LEF1 and BACH2 - is similarly stable. The mixed-lineage NBM block also persists, which is important: LODO robustness demonstrates reproducibility across the assembled datasets, but it does not convert a systematic technical or compositional difference into a cell-intrinsic biological effect.

MMRF_ImmuneAtlas is the dominant NDMM source with 206 of 233 NDMM pseudobulk replicates. Omitting it weakens or reverses IFNG, TIGIT, ISG15, FOXO1 and SPI1. These genes should therefore be treated as cohort-sensitive even when their full-data FDR is significant. In contrast, the strongest interferon genes and KLRK1/TNF are not dependent on the MMRF cohort for direction.

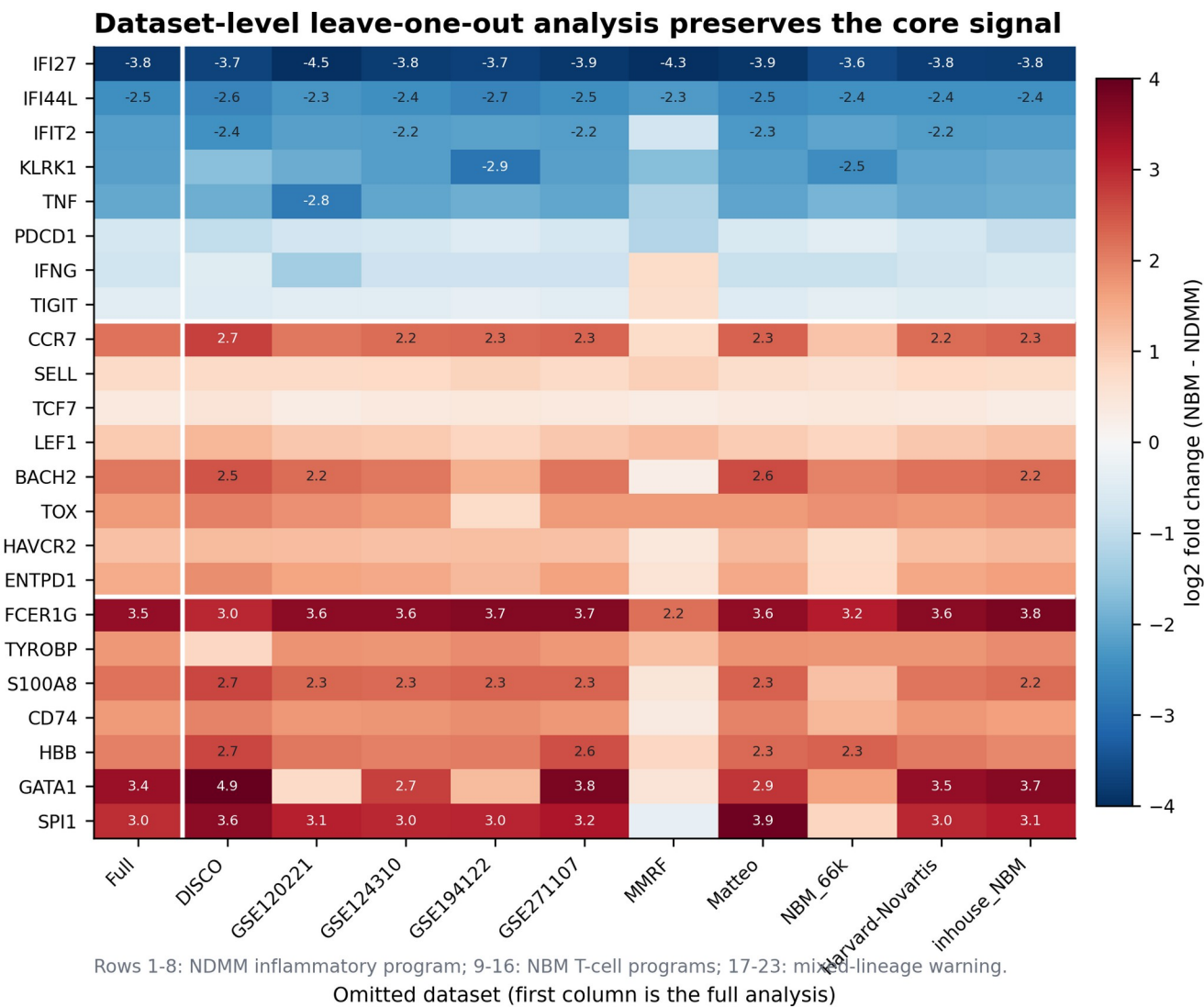


Figure 4. Full-data and leave-one-dataset-out log2 fold changes for selected genes. Blue indicates higher expression in NDMM; red indicates higher expression in NBM. The MMRF omission column identifies cohort-sensitive signals while preserving the main interferon block. Evidence: lodo_fold_results.csv

Robustness should be interpreted in two layers

What LODO supports	What LODO does not support
The effect is not abolished by removing any one represented dataset.	The group effect is independent of dataset.
The direction is reproduced across many dataset omissions.	The same disease effect would occur within a balanced dataset.
The signal is useful for prioritizing robust atlas-associated features.	Causality, antigen specificity or clinical translation.

4. The design is statistically estimable but biologically non-identifiable

The DESeq2 model is full rank and estimable as $\sim$ group with 286 residual degrees of freedom. That statement is mathematically correct for the assembled pseudobulk table. It does not mean the disease effect can be separated from dataset, because the ten datasets are group-exclusive: six contribute only NBM and four contribute only NDMM. The shared-dataset list is empty.

The imbalance is also extreme. MMRF_ImmuneAtlas contributes 206 NDMM replicates, while the largest NBM dataset contributes 21. The full model therefore estimates an atlas-level contrast between two collections of datasets. This is useful for discovery and hypothesis prioritization, especially with LODO sensitivity analysis, but it is not a substitute for within-dataset case-control inference or a formal meta-analysis of comparable cohorts.

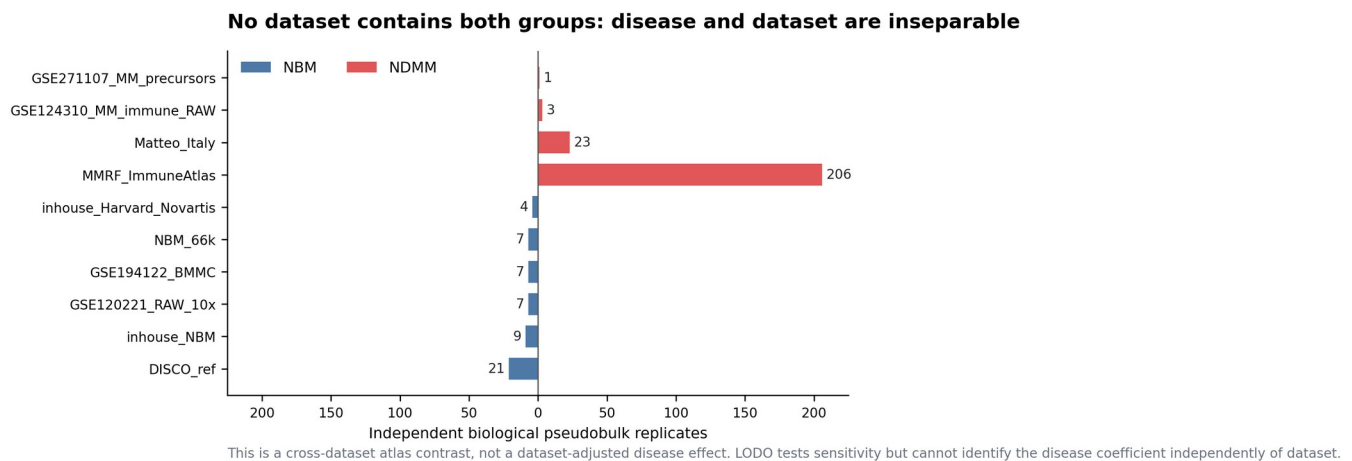


Figure 5. Independent pseudobulk replicate representation by dataset and group. No dataset contributes both NBM and NDMM samples, making dataset and disease group inseparable in the current design. Evidence: replicate_counts_by_dataset_group.csv; deseq2_design_assessment.json

Defensible wording for this result

Across the assembled atlas datasets, GZMB CD8 T cells associated with NDMM show a robust interferon-inflammatory program relative to cells associated with NBM. The signal persists under dataset-level leave-one-out analysis, but the current design cannot determine how much reflects disease biology versus systematic differences among datasets.

Language to avoid

- "NDMM causes" the observed expression program.
- "Dataset-adjusted" or "independent of cohort" disease effects.
- Formal claims of exhaustion, tumor specificity or impaired killing without orthogonal evidence.
- Treating LODO-conserved mixed-lineage genes as validated CD8 T-cell biology.

5. TF activity is lineage-sensitive and should not lead the story

Signed TF activity produced 14 significant resource-specific activities representing 13 regulators. All significant scores are positive in the NBM - NDMM orientation. SPI1 is the only regulator with directional consensus across both CollecTRI and DoRothEA. The remaining significant activities are single-resource signals, including SP1, JUN, CEBPB, NFkB, RUNX1, GATA1, TAL1 and ETS-family factors.

The consensus does not rescue a clean T-cell regulatory interpretation. SPI1 target support includes myeloid-lineage genes such as FCER1G, TYROBP, CSF1R, LYZ, MPO and related targets, while GATA1 and TAL1 are linked to the erythroid block. The TF results therefore reinforce the same QC concern exposed by gene-level differential expression: the NBM-associated feature universe contains lineage-sensitive signal that can dominate regulon scoring.

The TF node also reports 1,373 eligible genes absent from all signed regulons, 944 non-estimable resource-program-TF models, 264 resource-discordance instances and 12 significant activities supported by only one resource. These are not reasons to discard the analysis; they are reasons to subordinate TF claims to the more direct and coherent gene-level findings.

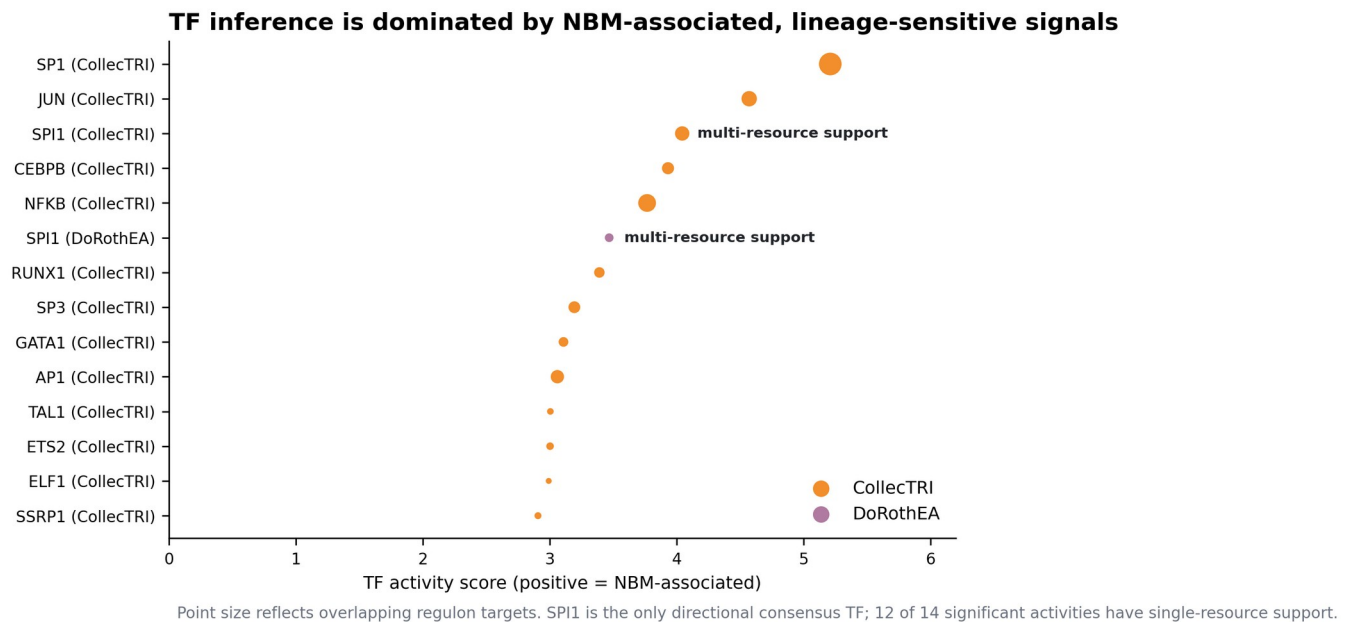


Figure 6. Significant signed TF activity scores. Point size reflects the number of overlapping regulon targets. SPI1 is the only directional consensus regulator; the remaining significant activities are single-resource and largely NBM-associated. Evidence: significant_tf_activity.tsv; tf_activity_consensus.tsv; qc_summary.json

TF conclusion

No robust NDMM-associated TF consensus is identified. SPI1 is the only cross-resource consensus signal, but its target structure and the accompanying GATA1/TAL1 activity make it more useful as a diagnostic of lineage-sensitive NBM signal than as a mechanistic CD8 T-cell regulator.

6. Integrated biological model

The most coherent interpretation is a remodeling of the broad GZMB CD8 compartment rather than a one-dimensional shift in cytotoxic strength. The NDMM-associated compartment is enriched for interferon response, inflammatory cytokine expression, NKG2D/KLRK1 and inhibitory receptors. The NBM-associated compartment contains both differentiation-preserving and chronic-stimulation programs, plus a strong mixed-lineage signal. These findings are compatible with disease-associated changes in the relative abundance of cytotoxic CD8 substates, exposure to distinct marrow cytokine environments, and systematic differences in cell recovery or annotation across cohorts.

Priority hypotheses

Priority	Hypothesis	Decisive test
1	NDMM marrow drives an interferon-high, TNF/IFNG-positive, checkpoint-adapted GZMB CD8 substate.	Validate within a dataset containing both NBM and NDMM; test IFN-response scores and PD-1/TIGIT protein.
2	The broad GZMB CD8 label contains distinct memory-like, cytotoxic and chronically stimulated substates whose proportions differ by group.	Recluster/subcluster the state and perform patient-level differential abundance plus within-substate pseudobulk DE.
3	The NBM SPI1/GATA1 block reflects ambient RNA, doublets or annotation/preprocessing differences rather than CD8 biology.	Apply ambient correction, doublet review and T-cell-restricted gene filters; rerun DE and TF activity.
4	KLRK1 and TNF are more reproducible NDMM markers than IFNG/TIGIT, which depend more strongly on MMRF.	Inspect per-dataset effects and validate in an independent cohort with balanced sampling.

Recommended next analysis sequence

- Create a harmonized subclustering of GZMB CD8 T cells and quantify memory-like, interferon-high, checkpoint-high and terminal effector substates per biological replicate.
- Repeat pseudobulk DE within each well-powered dataset that contains both groups, or assemble a formal random-effects meta-analysis from balanced cohorts. Dataset-exclusive contrasts should remain explicitly exploratory.
- Run ambient-RNA and doublet sensitivity analyses and remove myeloid/erythroid genes from the TF input universe before interpreting regulatory activity.
- Plot per-dataset effect sizes and pseudobulk expression for KLRK1, TNF, IFI27, IFI44L, IFIT2, PDCD1, IFNG and TIGIT. Separate robust core markers from MMRF-sensitive markers.
- Validate the prioritized program orthogonally with protein or functional measurements rather than relying on regulon scores alone.

Bottom line

The current bundle contains a real and biologically coherent NDMM-associated interferon/inflammatory signal. The strongest reportable result is not "thousands of genes changed" and not "SPI1 is activated." It is that the GZMB CD8 compartment associated with NDMM is qualitatively remodeled toward interferon response, inflammatory cytokines, KLRK1 and inhibitory receptors, while the NBM side is heterogeneous and partly contaminated by

lineage-sensitive signal. The finding is robust for discovery, but the design requires balanced validation before it becomes a disease-mechanistic claim.

Appendix: analysis and evidence provenance

Analysis design

Biological comparison	GZMB CD8 T cells; Normal Bone Marrow (NBM) - Newly Diagnosed Multiple Myeloma (NDMM).
Unit of inference	Independent biological pseudobulk replicate.
Input footprint	167,355 cells; 15,412 atlas genes; raw count source layers/counts.
Eligibility	Minimum 100 contributing cells per pseudobulk replicate.
Replicates	NBM 55; NDMM 233; 288 total across 10 datasets.
Differential expression	DESeq2 design ~ group; 5,213 genes retained and tested; Benjamini-Hochberg FDR.
Differential results	3,320 significant genes: 2,213 higher in NBM and 1,107 higher in NDMM.
Robustness	Ten dataset-level leave-one-out folds; all estimable; 2,395 features met the configured conservation rule.
TF activity	Signed regulon activity from CollecTRI and DoRothEA A-C using a documented ULM-style model over conserved gene statistics.
Design warning	No shared datasets between NBM and NDMM; group and dataset are not independently identifiable.

Primary evidence artifacts

- artifacts/deseq2/deseq2_results.tsv - full DESeq2 effect sizes, P values, FDR and direction.
- artifacts/deseq2/lodo_feature_summary.csv and lodo_fold_results.csv - gene-level robustness across omitted datasets.
- design/replicate_counts_by_dataset_group.csv and deseq2_design_assessment.json - cohort structure and identifiability.
- artifacts/tf_activity/significant_tf_activity.tsv, tf_activity_consensus.tsv and qc_summary.json - regulator-level inference and QC.
- evidence_bundle.json - bundle identity, provenance, warnings and deterministic evidence references.

Interpretation note

Gene-program labels in this report are curated biological interpretations of the observed marker pattern. They are not formal pathway-enrichment results because no enrichment artifact was present in the evidence bundle. TF activity scores are model-based associations with signed regulon targets and are not direct measurements of TF protein activity, binding, phosphorylation or causality.

All report values were derived from the supplied public evidence bundle. The report preserves the original contrast orientation and warning structure while separating robust biological signal from design and lineage-sensitivity limitations.